

JEE MAIN REVISION MASTER: CIRCULAR MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 7 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Circular Motion — Core Principles & Governing Laws	6 PYQs	Core Principle
Moderate	Circular Motion — Standard Calculations & Problem Archetypes	3 PYQs	Quantitative Analysis
Standard	Circular Motion — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Circular Motion — Core Principles & Governing Laws	10ms^{-1} , $r=10\text{m}$, 2ms^{-1}
Circular Motion — Standard Calculations & Problem Archetypes	$(x-2)^2+y^2=25$, 2ms^{-1}

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Circular Motion — Core Principles & Governing Laws (Moderate Yield — 6 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Circular Motion.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Circular Motion.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Circular Motion PYQs.

Representative Source PYQs:

- **[2025 Shift] Q#1 (MCQ):** "A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2+y^2=25$, where x and y are in meter. The ..." **[Ans: N/A]**
- **[2023 Shift] Q#3 (MCQ):** "A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?..." **[Ans: N/A]**
- **[2022 Shift] Q#4 (MCQ):** "A car is moving in a circular horizontal track of radius 10m with a constant speed of 10ms^{-1} . A bob is suspended from the roof of th..." **[Ans: N/A]**

3.2 Circular Motion — Standard Calculations & Problem Archetypes (Moderate Yield — 3 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Circular Motion.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- **[2025 Shift] Q#1 (MCQ):** "A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The ..." **[Ans: N/A]**
- **[2024 Shift] Q#2 (MCQ):** "An athlete runs along a circular track of diameter 80 m. The distance travelled and the magnitude of displacement of the athlete when he co..." **[Ans: N/A]**
- **[2023 Shift] Q#3 (MCQ):** "A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?..." **[Ans: N/A]**

3.3 Circular Motion — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Circular Motion.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- **[2023 Shift] Q#3 (MCQ):** "A particle is in uniform circular motion, related to one complete revolution of the particle, which among the statements is incorrect?..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

A particle is in uniform circular motion. The equation of its trajectory is given by $(x-2)^2 + y^2 = 25$, where x and y are in meter. The speed of the particle is 2ms^{-1} , when the particle attains the lowest 'y' co-ordinate, the acceleration of the particle is (in ms^{-2})

- (A) 0.4j
- (B) 0.8i
- (C) 0.8j
- (D) 0.4i

Correct Answer: None

Solution Analysis: To determine the acceleration of a particle in uniform circular motion, we start with the equation of the circle: $(x-2)^2 + y^2 = 25$. This equation represents a circle with its center at $(2,0)$ and a radius r of 5 meters, derived from: $r^2 = 25 \Rightarrow r = 5$. In uniform circular motion, the centripetal acceleration a_c is given by the formula: $a_c = \frac{v^2}{r}$. Substituting the given speed $v = 2\text{m/s}$ and the radius $r = 5\text{m}$: $a_c = \frac{2^2}{5} = 0.8\text{m/s}^2$. When the particle is at the lowest y-coordinate, it is at the point $(2, -5)$, assumin

Benchmark Problem 2: KCET 2024 (Q#2)

An athlete runs along a circular track of diameter 80 m . The distance travelled and the magnitude of displacement of the athlete when he covers $\frac{3}{4}$ th of the circle is (in m)

- (A) $60\pi, 40\sqrt{2}$
- (B) $40\pi, 60\sqrt{2}$
- (C) $120\pi, 80\sqrt{2}$
- (D) $80\pi, 120\sqrt{2}$

Correct Answer: None

Solution Analysis: Given, diameter, $d=80\text{m}$ $\therefore r=40\text{m}$ Distance travelled after completion of $(\frac{3}{4})$ revolution $= (\frac{3}{4}) \times 2\pi r = (\frac{3\pi}{2}) \times 40 = 60\pi$ m Displacement $= \sqrt{(r^2) + r^2} = r\sqrt{2} = 40\sqrt{2}\text{m}$